

1. Evaluate the integral. $\int_1^{\infty} x e^{-x^2} dx$
2. Evaluate the integral. $\int_0^1 \frac{1}{(1-x)^2} dx$
3. Does the sequence $a_n = \frac{8^{n+1} + 2^{n-2} + 5^n}{8^n}$ converge or diverge? If it converges, what does it converge to?
4. Consider the infinite series

$$21 - 15 + \frac{75}{7} - \frac{375}{49} + \frac{1875}{343} - \dots$$

Write this series in the form $\sum_{n=1}^{\infty} a_n$. Does this series converge? If so, find the sum.

5. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=1}^{\infty} \frac{n^n}{(-4)^n}$$

6. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=2}^{\infty} \frac{3}{n(\ln(n))^2}$$

7. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=1}^{\infty} (-1)^n \frac{5n}{4n+7}$$

8. Determine whether the series is convergent or divergent. Clearly state which convergence test you used.

$$\sum_{n=1}^{\infty} \frac{6n}{5n^2 + 4n + 7}$$

9. Determine whether the series is absolutely convergent, conditionally convergent, or divergent. Clearly state which convergence test(s) you used.

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{2^n}$$

10. Determine whether the series is absolutely convergent, conditionally convergent, or divergent. Clearly state which convergence test(s) you used.

$$\sum_{n=2}^{\infty} (-1)^{n+1} \frac{n^2}{n^3 - 3}$$

11. For the following power series, find the radius and interval of convergence.

$$\sum_{n=0}^{\infty} \frac{3^n}{(n+1)^3} (2x-1)^n$$

12. Find the radius and interval of convergence.

$$\sum_{n=2}^{\infty} (-1)^n \frac{x^n}{4^n \ln n}$$

13. Find the radius and interval of convergence.

$$\sum_{n=1}^{\infty} \frac{3^n (x+4)^n}{\sqrt{n}}$$

14. Find the power series representation of the function and determine the radius of convergence.

$$f(x) = \frac{x}{9+x^2}$$

15. Find the power series representation of the function and determine the radius of convergence.

$$f(x) = \frac{x}{(1+4x)^2}$$

16. Determine the Maclaurin series for $f(x) = 4x^4 \cos(\pi x^5)$.

17. Find the **Taylor series** for $f(x) = \ln(x)$ centered at $a = 2$. Express your answer in summation notation.

18. How many terms are needed to approximate the sum below to four decimal places?

$$\sum_{n=1}^{\infty} \frac{(-1)^n \cdot n}{8^n}$$

19. How many terms are needed to approximate the sum below to ensure the error is less than 0.0000001 using the Integral Test Approximation?

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

20. Find the radius and ‘disk’ of convergence for the Complex Power Series below:

$$\sum_{n=1}^{\infty} 2^{2n} \left(z - \frac{1}{2}\right)^n$$